ONLINE MASTERS IN **DATA SCIENCE**

DSC 258 – NATURAL LANGUAGE PROCESSING

RNN LANGUAGE MODELS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

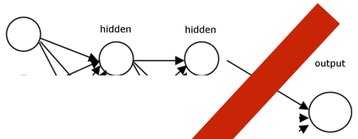
HALICIOĞLU DATA SCIENCE INSTITUTE

Feed-forward vs. Recurrent

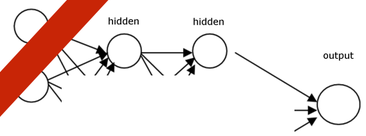
What if we *now* remove the Markov assumption?

$$p(x_i \mid x_1, \ldots, x_{i-1})$$

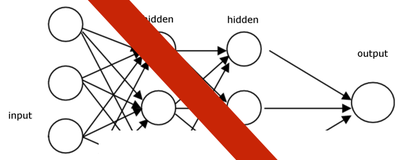
Context size 1:



Context size 2:



Context size 3:



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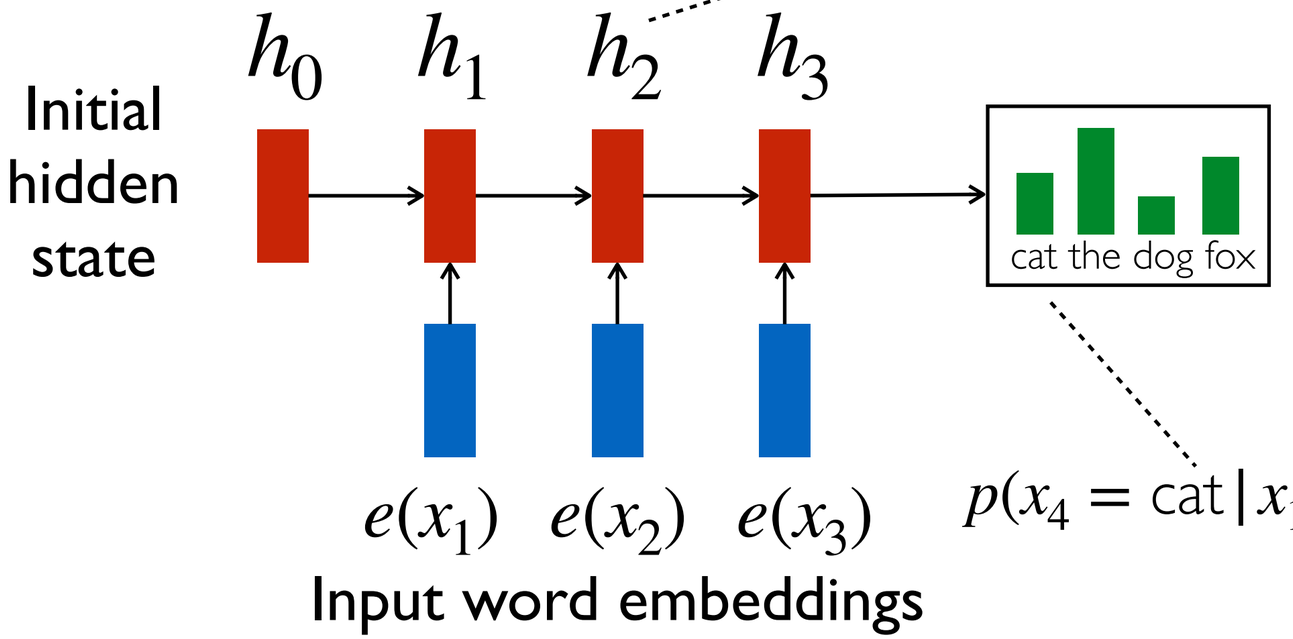
Recurrent Neural Network LM

$$h_i = \sigma(W^\top h_{i-1} + U^\top x_i)$$



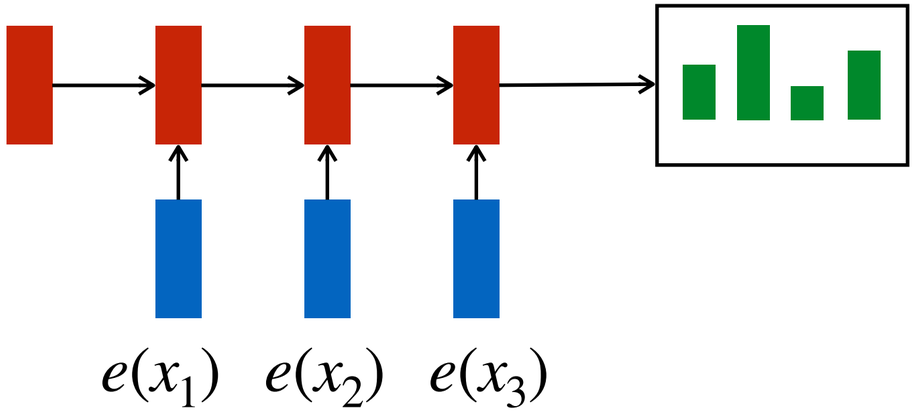
$$p(x_4 \mid x_1, x_2, x_3) = \text{softmax}(V^\top h_3)$$

$$p(x_4 = \text{cat} \mid x_1, x_2, x_3) = \frac{\exp(v(\text{cat}) \cdot h_3)}{\sum_x \exp(v(x) \cdot h_3)}$$

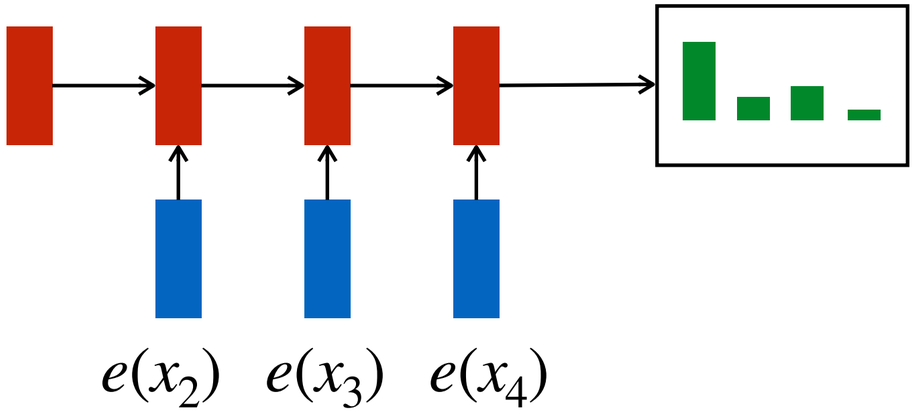
Input embedding matrix: $E \in \mathbb{R}^{d \times m}$

Output embedding matrix: $V \in \mathbb{R}^{d \times m}$

Recurrent Neural Network LM



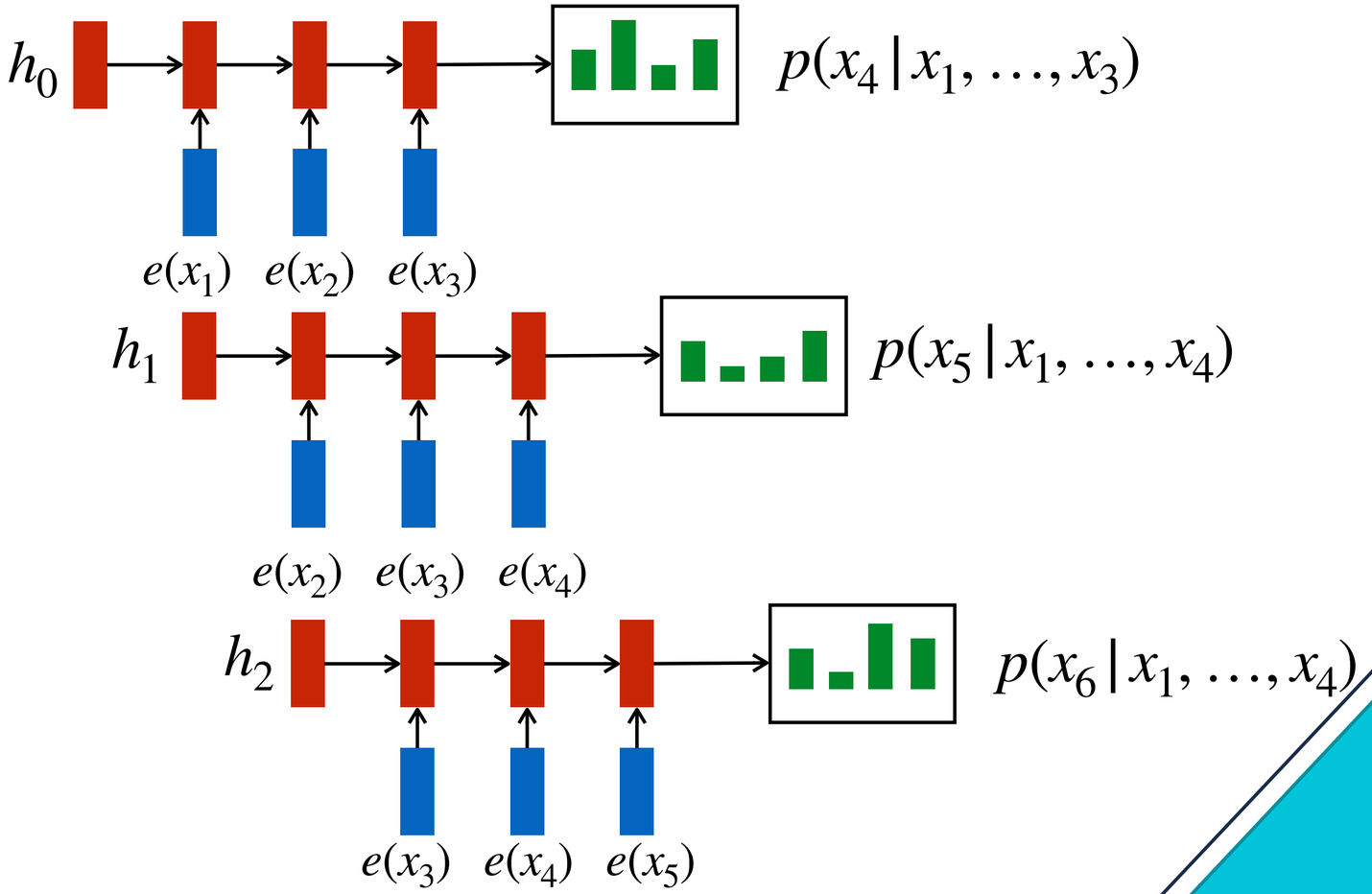
$$p(x_4 \mid x_1, x_2, x_3)$$



$$p(x_5 \mid x_2, x_3, x_4)$$

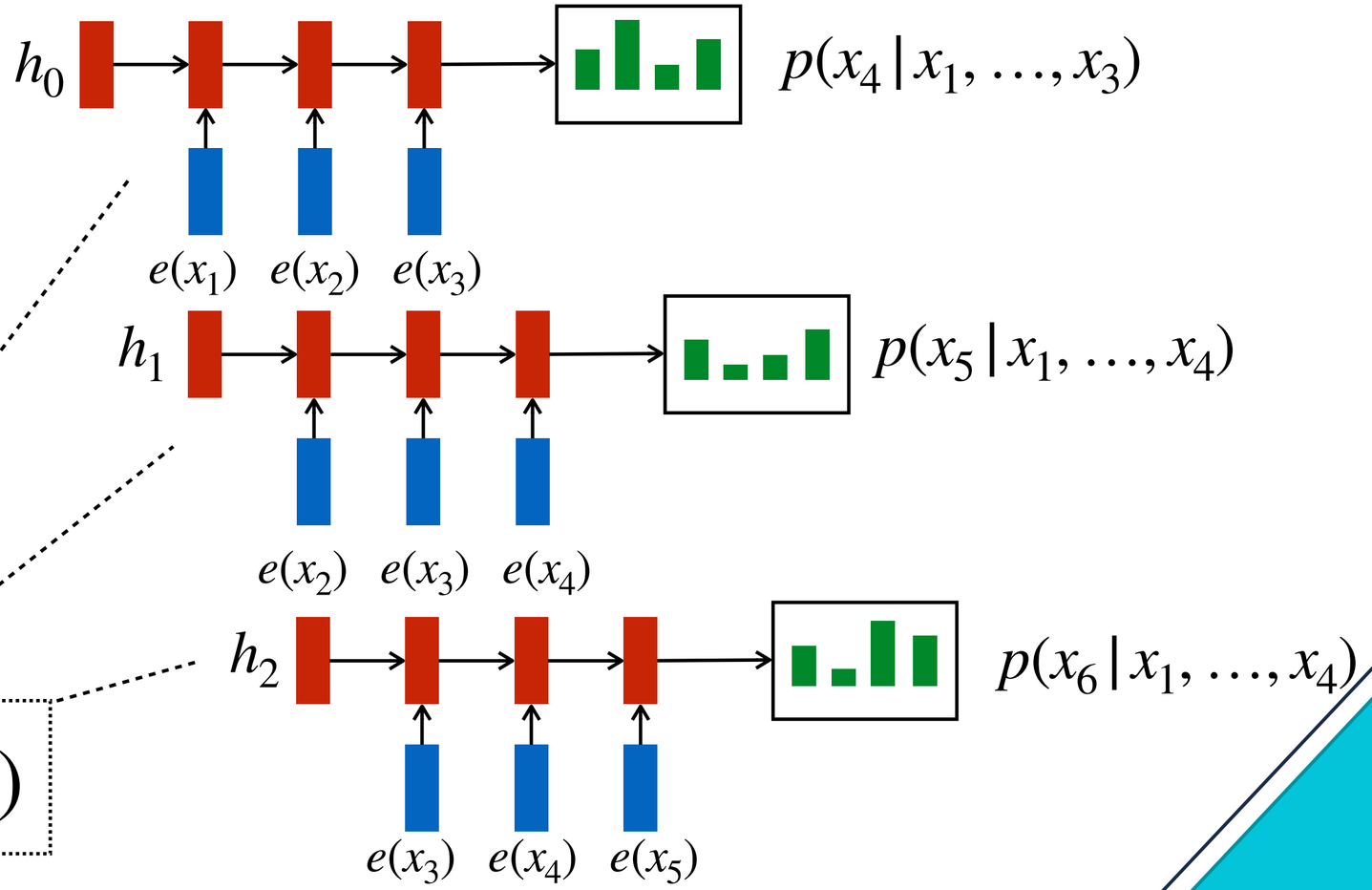
Recurrent Neural Network LM

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$



Recurrent Neural Network LM

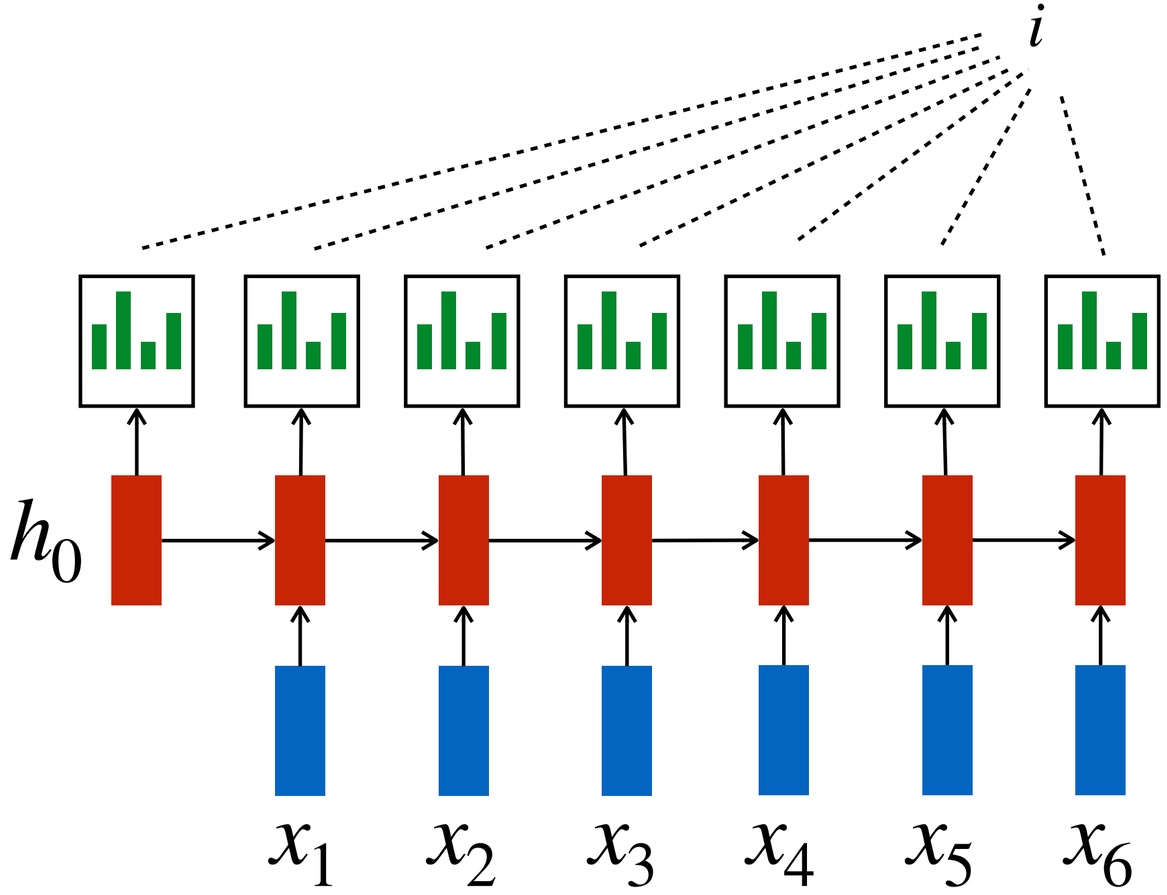
$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$



$$\theta = (E, V, U, W, h_0)$$

Recurrent Neural Network LM

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$

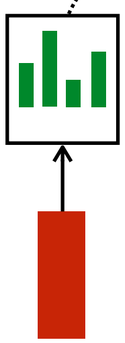


$$\theta = (E, V, U, W, h_0)$$

Sampling

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$

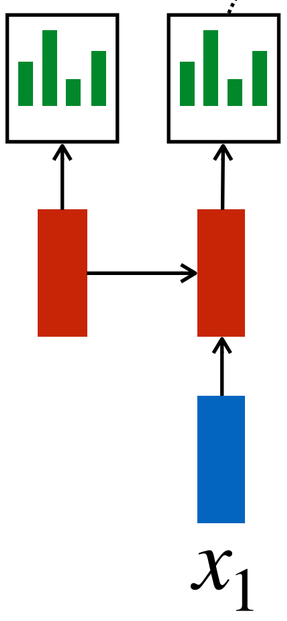
$$x_1 \sim p(x_1)$$



Sampling

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$

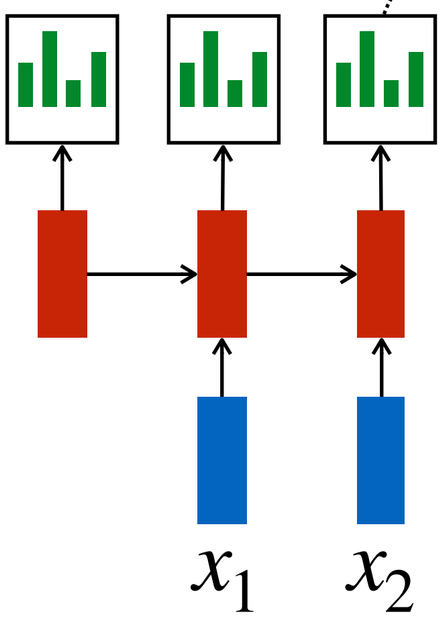
$$x_2 \sim p(x_2 \mid x_1)$$



Sampling

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$

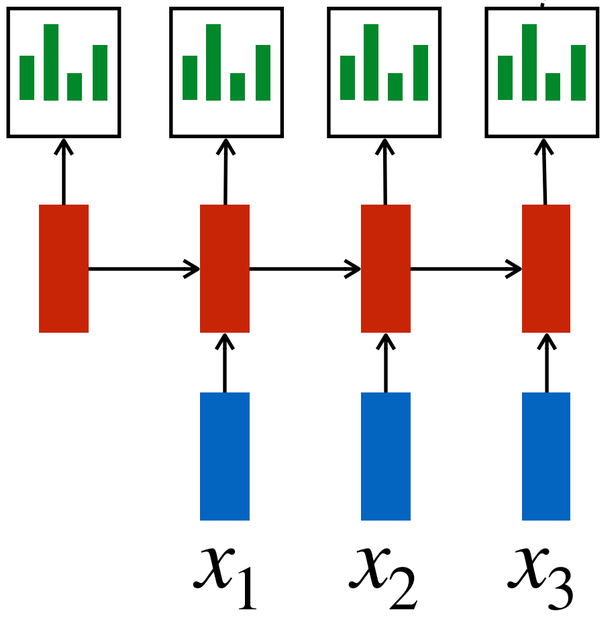
$$x_3 \sim p(x_3 \mid x_1, x_2)$$



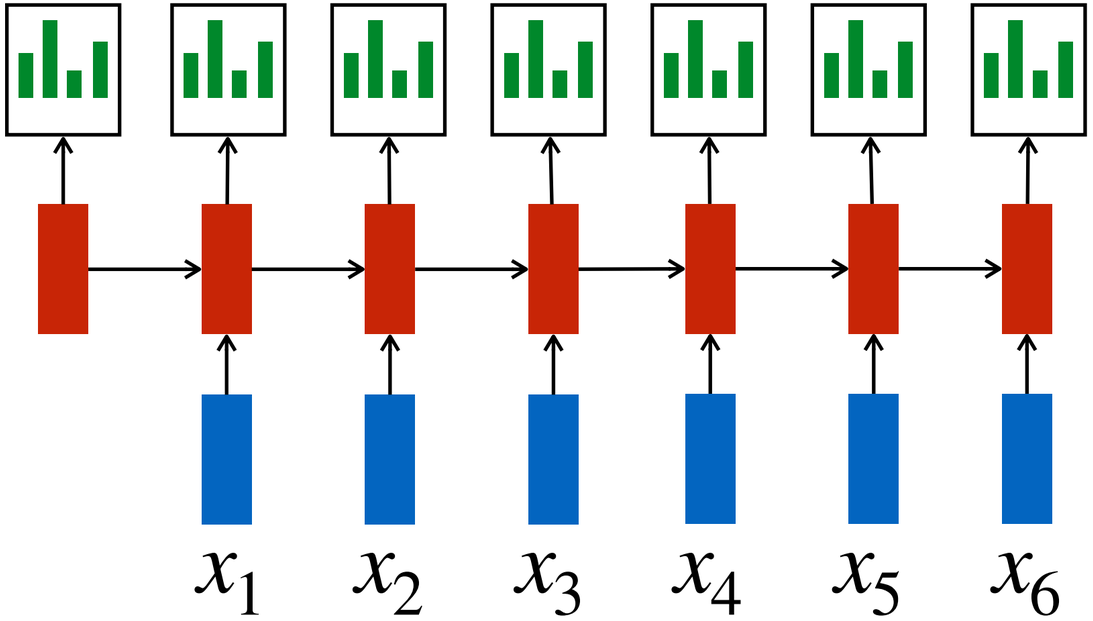
Sampling

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$

$$x_4 \sim p(x_4 \mid x_1, x_2, x_3)$$



Sampling

$$p(x_1, \ldots, x_n) = \prod_i p(x_i \mid x_1, \ldots, x_{i-1})$$

$$x_i \sim p(x_i \mid x_1, \ldots, x_{i-1})$$